Wind turbine project
Part I: Wave Dynamics
A. Characterizing the study site mean wave conditions
October 9, 2026

Part I of the wind turbine project spans 2 classes and consists of analyzing the wave conditions at your selected study sites. Last week you already began to select your study site, using criteria based on a wind speed threshold, water depth, and distance from the coast. Over the next 2 classes, you will: (A) characterize the mean wave conditions and seasonal variability, and (B) estimate the extreme wave conditions. You will be asked to write up a concise report of your results of Parts I. A & B of the project (due on 23/10/2026). Some of this information will be needed in Parts II and III of your project to model wave propagation and to estimate the impacts of waves on an offshore wind turbine (OWT), during both normal operating conditions and extreme events. For the report concerning Part I of the project, please respond to the questions in I.A. (listed below) and I.B. (provided in next week's class).

A. Characterizing the mean wave conditions and the seasonal variability

The objective during this class is to characterize the mean wave conditions, as well as the seasonal and interannual variability. Start by downloading the longest wave time series (20-30 years, ideally) that you can find at your study site. If you have a choice betewen both observations (a wave buoy) and model simulation results, you can explore the results of both of these analyses if you have time.

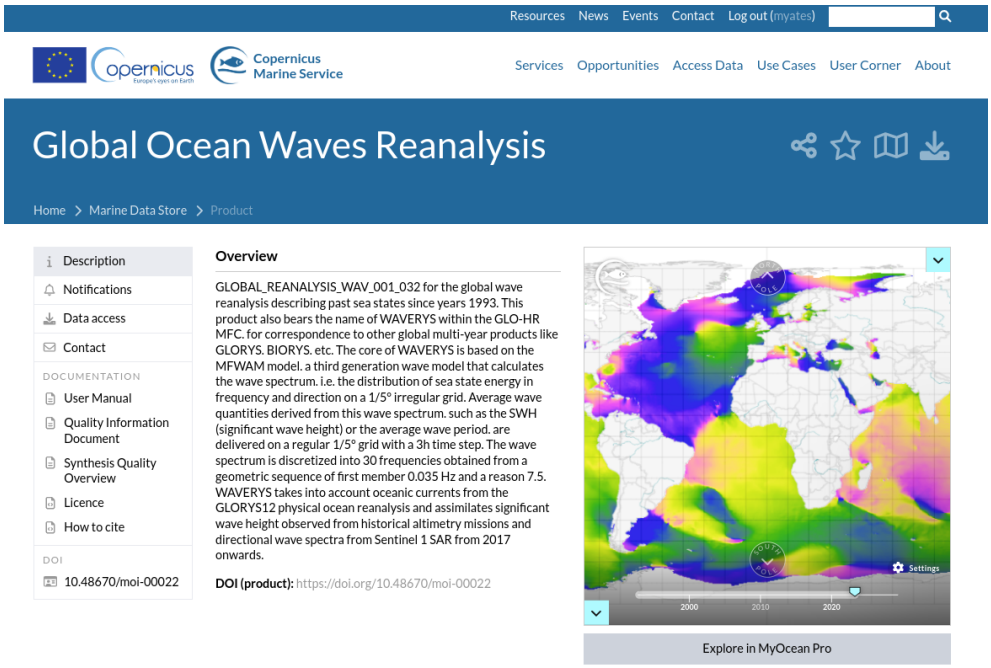


FIGURE 1. Example available Global Ocean Waves Reanalysis data set from the Copernicus Marine Service website.

To verify the water depth and create a **bathymetric map** for your study site, you can extract bathymetric data from the EMODNET database via their website (https://www.emodnet-bathymetry.eu/data-products). You can visualize profiles using the 'Bathymetry Viewing and Downloading Service'. You can then use the **copernicusmarine** python toolbox to load and analyze the wave data.

1. Mean wave conditions. Please download the variables corresponding to the significant wave height, mean wave period (T_{m02}), and mean wave direction. Then plot the time series of these variables and calculate the mean significant wave height, period, and direction over the entire available time series.

Q: Do any trends in the wave height, period, or direction exist over the 26 year time period? Do you expect there to be changes in the mean conditions during the 30-year lifetime of the wind turbine? If so, why?

2. Most common wave conditions. Please make a 2D historgram of the significant wave height versus the mean wave period (e.g. like the example shown in class 1) and identify the most commonly observed wave conditions (H_{m0},T_{m02}) at your study site. Then plot the wave rose to identify the most common wave incidence direction(s).

Q: What is the water depth at this location? Indicate on the histrogram for what wave conditions the waves are considered deep water waves? linear waves?

3. Seasonal variability. To evaluate the temporal variability, calculate and plot these three quantities as a function of the month of the year (e.g. calculate the mean for the month of January, February, etc.).

Q: Do you observe any seasonal trends in wave height, period, or direction? If so, why?

You are encouraged to work together in groups of 2-3 people to do these analyses, but you are asked to write up individual reports (short description of the results). Please include the requested figures, question responses, and any additional relevant data interpretation.